

Production Dashboard — Detailed Design Document (詳細設計書)

003_DD — implements 003_BD (SCR-003), requirements REQ-028–REQ-042.

Document control (改版履歴)

Field	Value
Document ID	003_DD
Category	UI + API
System name	ProductionManagementAI
Subsystem name	Production orders
Work item	WI-004
Implements	003_BD revision 3
Created by	Claude (for ThanhTN)
Created date	2026-09-22
Last updated by	Claude (for ThanhTN)
Last updated date	2026-09-23

Version	Date	Author	Revision content
1	2026-09-22	Claude (for ThanhTN)	Initial creation
2	2026-09-22	Claude (for ThanhTN)	Rendered mockup published and linked
3	2026-09-22	Claude (for ThanhTN)	Plan revision 2 (mockup review): navbar in the shared header and the navigation guard (DEC-016, DEC-022), health indicator and its endpoint (DEC-017, DEC-019, DEC-020), chart maximize (DEC-018); page actions removed; test viewpoints TC-222–TC-227 added, TC-201/TC-202 updated
4	2026-09-22	Claude (for ThanhTN)	Icons (DEC-023): <code>lucide-react</code> dependency and the <code>components/icons.ts</code> import point; mockup v3

Version	Date	Author	Revision content
5	2026-09-23	Claude (for ThanhTN)	Diagrams redrawn for the revised DD template (RFC 0009): the layout sketch as an SVG wireframe under <code>wireframes/</code> ; the screen states also drawn as a Mermaid state diagram above the table. No field, rule, API or processing changes
6	2026-09-23	Claude (for ThanhTN)	WI-005: the UI is Japanese. The message catalog lists the Japanese text with an English gloss (WI-005 DEC-003); formatter rows (件, 個, 日, YYYY/MM/DD , M/D) and names match the code; health texts, wireframes and mockup updated. Corrected: the <code>ready</code> state announces the snapshot-time line, not a separate "Dashboard updated" text. A Japanese edition of the mockup was added (<code>mockups/003_DD_screen-c-mockup.ja.html</code>). No field, rule, API or processing changes

Overview and reference documents (概要・目次)

Field	Value
File / component name	<code>DashboardPage.tsx</code> and its child components; <code>DashboardController.Get</code>
Overview	SCR-003, the production dashboard at <code>/</code> : eight read-only widgets rendered from one server snapshot, a server/database health indicator, and charts that maximize to the full window; replacing the placeholder home page. Also the shared header's navbar and navigation guard used by every screen

Module / method / processing index

No	Name	Overview	Notes
1	<code>DashboardPage</code>	Route component: role gate, load, state selection, page actions	Module design §1
2	<code>StatusTiles</code>	Items 7–11	§2
3	<code>DeliveryTiles</code>	Items 12–15	§2
4	<code>AttentionList</code>	Items 16–19	§3
5	<code>TopProducts</code>	Item 22	§4
6	<code>BarChart</code>	Items 20–21 and 23–24: SVG bars plus table equivalent	§5
7	<code>dashboardFormat.ts</code>	Pure formatters for M-13, M-14, M-16–M-19	§6
8	<code>dashboardApi.ts</code> — <code>getDashboard</code>	Typed client wrapper	§7
9	<code>DashboardController.Get</code>	Request boundary	§8

No	Name	Overview	Notes
10	<code>DashboardService</code> , <code>DashboardWindow</code> , <code>IDashboardReader</code> , <code>DashboardMapper</code> , <code>IPlantClock</code> additions	Server-side snapshot	Pointer only — designed in 003_DD-FN
11	<code>ProductionOrder</code> entity — completion time	REQ-033	Pointer only — designed in 001_DD v4 module 1 step 5
12	<code>AppNavbar</code> (in <code>AppHeader</code>)	The application navbar on every authenticated screen	\$10
13	<code>NavigationGuard</code> + <code>GuardedLink</code>	Lets an edited SCR-001 form intercept in-app links (DEC-022)	\$11
14	<code>HealthIndicator</code> + <code>useSystemHealth</code>	Item 26: status, polling, visibility pause	\$12
15	<code>ChartDialog</code>	Items 27–29: maximized chart view	\$13
16	<code>SystemController.Health</code> , <code>SystemHealthService</code> , <code>IDatabasePing</code> , cookie renewal rule	Health endpoint and the no-renew rule	Pointer only — designed in 003_DD-FN \$6–\$7

Reference documents

No	Document	Purpose / use	Notes
1	003_BD	Items, metric definitions D-01–D-09, mappings M-11–M-19, events E-20–E-23	Parent
2	003_DB	Column, constraints, queries Q1–Q6, snapshot, seed, migrations	Data layer
3	003_DD-API	Endpoint contract	Companion
4	003_DD-FN	Service, window, reader, mapper, plant-clock additions	Companion
5	003_DD-SPD	Step-by-step processing per component	Companion
6	001_DD set (v4), 001_BD v6	Completion stamping on <code>InProgress</code> → <code>Completed</code> ; message catalog, <code>apiClient</code> , <code>MessageBanner</code> , plant clock	Screen A
7	002_DD set, 002_DB	<code>ProductSummary</code> , status labels, test conventions; seed that 003_DB extends	Screen B
8	0001_ADR, 0002_ADR	Layered backend; cookie auth and roles	

Referenced by

No	Document	Purpose / use	Notes
1	003_DD-API, 003_DD-FN, 003_DD-SPD	Companions elaborating this DD	
2	work-items/WI-004/test-plan.md (TP-004)	Test viewpoints below become TC-2##	Written in plan revision 2

Component / file organization

X-1. Path structure

No	Path / namespace	Purpose	Notes
1	src/frontend/src/features/dashboard/	New feature folder	DashboardPage.tsx , StatusTiles.tsx , DeliveryTiles.tsx , AttentionList.tsx , TopProducts.tsx , BarChart.tsx , dashboardFormat.ts , dashboardApi.ts , types.ts
2	src/frontend/src/App.tsx	Route / → DashboardPage ; the inline HomePage is deleted; the authenticated routes are wrapped in NavigationGuardProvider	DEC-005, DEC-022
2a	src/frontend/src/components/	AppHeader.tsx (extended), AppNavbar.tsx , GuardedLink.tsx	Shared by every screen (DEC-016)
2b	src/frontend/src/lib/navigationGuard.tsx	NavigationGuardProvider , useNavigationGuard	DEC-022
2c	src/frontend/src/features/dashboard/	Also HealthIndicator.tsx , useSystemHealth.ts , ChartDialog.tsx , systemApi.ts	

No	Path / namespace	Purpose	Notes
2e	src/frontend/src/components/icons.ts	The only module importing <code>lucide-react</code> ; re-exports the icons of 003_BD M-21 under role names (<code>NavDashboardIcon</code> , <code>StatusCompletedIcon</code> , ...), so a glyph changes in one place	DEC-023
2f	src/frontend/package.json	<code>lucide-react</code> pinned at an exact version (1.47.0 at design time; confirmed when installed)	DEC-023
2d	src/frontend/src/features/production-orders/ProductionOrderForm.tsx , ProductionOrderPage.tsx , ProductionOrderListPage.tsx	The form registers with the guard; breadcrumb links become <code>GuardedLink</code>	DEC-022
3	src/frontend/src/features/production-orders/messages.ts	The single message catalog, extended with MSG-E021 and MSG-I005–MSG-I008	Not a second catalog
4	src/backend/ProductionManagementAI.Api/Controllers/DashboardController.cs	New controller	
5	src/backend/ProductionManagementAI.Application/Dashboard/	<code>DashboardService</code> , <code>DashboardWindow</code> , <code>DashboardMapper</code> , <code>IDashboardReader</code> , <code>DashboardContracts</code>	New namespace
6	src/backend/ProductionManagementAI.Infrastructure/Dashboard/DashboardReader.cs	Raw-SQL reader	
6a	src/backend/ProductionManagementAI.Api/Controllers/SystemController.cs , Application/Health/SystemHealthService.cs (+ <code>IDatabasePing</code>), Infrastructure/Health/DatabasePing.cs (namespace <code>Health</code> , WI-004 DEC-024)	Health endpoint	003_DD-FN \$6
6b	src/backend/ProductionManagementAI.Infrastructure/DependencyInjection.cs	Cookie <code>Events.OnCheckSlidingExpiration</code> : no renewal for the health path	003_DD-FN \$7, DEC-019
7	src/backend/ProductionManagementAI.Application/ProductionOrders/Ports.cs , Infrastructure/ProductionOrders/PlantClock.cs	<code>IPlantClock</code> additions	003_DD-FN \$5
8	src/backend/ProductionManagementAI.Domain/ProductionOrders/ProductionOrder.cs , Infrastructure/ProductionOrders/ProductionOrderConfigurations.cs	<code>CompletedAtUtc</code> property, mapping and check constraints	001_DD v4, 003_DB
9	src/backend/ProductionManagementAI.Infrastructure/Migrations/	003_DB's three migrations	

X-2. Shared/common components used

No	Name	Purpose	Notes
1	apiClient	Fetch wrapper with 401 → /login	WI-001
2	AppHeader , ProtectedRoute	Common header (now with AppNavbar) and route guard	DEC-016
3	MessageBanner	Error banner with its live region	001_DD
4	messages.ts , statusLabels	Message catalog and status labels	001_DD, 002_DD

X-3. Feature-level components used — modules 2–6 above.

X-4. External APIs used — see "APIs used" below.

X-5. Responsive composition — one set of components with responsive Tailwind classes. The widget grid is one column below `sm`, two columns at `lg` for the paired widgets (attention list with top products; workload with trend). Tile rows use `grid-cols-2` below `sm` and `grid-cols-5` / `grid-cols-4` above. Two components change markup rather than only layout: `AttentionList` (table ↔ cards, as `ProductionOrderTable` does), and `BarChart`, which keeps a minimum width inside a horizontally scrolling card on SP.

Companion design documents

No	Document	Type	Covers
1	003_DD-API_ダッシュボード.md	api-design	<code>GET /api/dashboard</code> (API-DSH-01) and <code>GET /api/system/health</code> (API-SYS-01): response catalogs, error codes, caching
2	003_DD-FN_ダッシュボード.md	function-design	<code>DashboardService.GetSnapshotAsync</code> , <code>DashboardWindow.For</code> , <code>IDashboardReader.ReadAsync</code> , <code>DashboardMapper.ToResponse</code> , <code>IPlantClock</code> additions, <code>SystemHealthService.CheckAsync</code> , the session-renewal rule, observability
3	003_DD-SPD_ダッシュボード.md	screen-processing-design	P-20 to P-29 plus the two controller request boundaries

Task / design index

No	Category	File-level task	Function/process-level task	Item-level task	Confirmed	Issue	Reviewer	Reworked	Date	Notes
1	basic-design	003_BD	SCR-003	—	yes	—	ThanhTN	no	2026-09-22	Approved ("003_BD is approved, move on to 003_DB")

No	Category	File-level task	Function/process-level task	Item-level task	Confirmed	Issue	Reviewer	Reworked	Date	Notes
2	database-design	003_DB	Column, queries, seed, migrations	—	yes	—	ThanhTN	no	2026-09-22	Approved ("003_DB is approved, move on to the DD")
3	basic/detailed-design	001_BD v6, 001_DD v4, 001_DD-FN v3, 001_DD-API v2	Completion stamping	—	yes	—	ThanhTN	no	2026-09-22	Approved with 003_DB
4	detailed-design	003_DD	Modules §1–§8, states, mockup	—	no	—	ThanhTN	no	2026-09-22	This document
5	api-design	003_DD-API	GET /api/dashboard	Response catalog	no	—	ThanhTN	no	2026-09-22	
6	function-design	003_DD-FN	Service, window, reader, mapper	Calendar rules, rounding	no	—	ThanhTN	no	2026-09-22	
7	screen-processing-design	003_DD-SPD	P-20–P-25	Chart geometry, response handling	no	—	ThanhTN	no	2026-09-22	

Module design

1. DashboardPage

Field	Value
Description	Route component for <code>/</code> : renders the page heading and actions, loads the snapshot once, and chooses among <code>loading</code> , <code>ready</code> , <code>error</code> and <code>forbidden</code>
Return type	JSX element
Created by / date	Claude / 2026-09-22
Last modified by / date	—

Preconditions: rendered inside `ProtectedRoute` , so a session exists.

Not applicable — plain component, no rule/validator fields.

Arguments: none (route component).

Dependencies

No	Module / component	Overview	Notes
1	<code>getDashboard</code>	Data	\$7
2	Modules 2–6, <code>MessageBanner</code>	Children	
3	<code>useAuth</code>	Role gate (UX only; the server is authoritative)	WI-001
4	<code>HealthIndicator</code>	Item 26, rendered beside the heading	\$12

Processing overview: 003_DD-SPD §1 (P-20) and §2 (P-21). Every widget receives its slice of the same snapshot object; none fetches on its own.

Processing flow: 003_DD-SPD §1, §2.

Return value

Type	Name	Description
JSX	—	Heading and health indicator always (the indicator is hidden in <code>forbidden</code>); below them one of <code>loading</code> , <code>ready</code> , <code>error</code> , <code>forbidden</code>

2. `StatusTiles` and `DeliveryTiles`

Field	Value
Description	The two tile rows (items 7–11, 12–15)
Return type	JSX element
Created by / date	Claude / 2026-09-22
Last modified by / date	—

Preconditions: `ready` state.

Not applicable — presentational.

Arguments

No	Type	Name	Description
1	<code>StatusCounts</code>	<code>counts</code>	<code>StatusTiles</code>
2	<code>CountAndQuantityWithFrom</code> ×2, <code>OnTime</code> , <code>LeadTime</code>	<code>week</code> , <code>month</code> , <code>onTime</code> , <code>leadTime</code>	<code>DeliveryTiles</code>

Dependencies: `dashboardFormat` (§6), `statusLabels` .

Processing overview: 003_DD-SPD §3 (P-22). Each tile is a list item with its label first in reading order and its value second, so a screen reader announces "Draft, 35" rather than a bare number.

Return value: JSX.

3. `AttentionList`

Field	Value
Description	Overdue and due-soon groups (items 16–19)
Return type	JSX element
Created by / date	Claude / 2026-09-22
Last modified by / date	—

Preconditions: `ready` state.

Not applicable — presentational.

Arguments

No	Type	Name	Description
1	<code>OrderGroup</code>	<code>overdue</code>	D-01
2	<code>OrderGroup</code>	<code>dueSoon</code>	D-02

Dependencies: `statusLabels` , `dashboardFormat.formatQuantity` .

Processing overview: 003_DD-SPD §4 (P-23). No cell is a link and no row has a handler (DEC-006). The status badge reuses Screen B's badge styling (text label, never color alone).

Return value: JSX.

4. TopProducts

Field	Value
Description	Ranked list of up to 10 products by open quantity (item 22)
Return type	JSX element
Created by / date	Claude / 2026-09-22
Last modified by / date	—

Preconditions: ready state.

Not applicable — presentational.

Arguments

No	Type	Name	Description
1	TopProduct[]	products	D-04, in rank order

Dependencies: dashboardFormat.formatQuantity.

Processing overview: 003_DD-SPD §5 (P-24).

Return value: JSX (<o1>).

5. BarChart

Field	Value
Description	Reusable inline-SVG bar chart with a printed value per bar, an accessible summary and a 「表で表示」 (View as table) disclosure (DEC-010)
Return type	JSX element
Created by / date	Claude / 2026-09-22
Last modified by / date	—

Preconditions: none — it renders any number of non-negative values, including all zero.

Not applicable — presentational.

Arguments

No	Type	Name	Description
1	string	title	The widget heading (<h2>)
2	{ label: string; value: number; emphasis?: boolean }[]	bars	One entry per bar, in display order; emphasis marks the overdue and current-week bars
3	string	summary	The SVG's accessible name
4	string[] / string[][]	tableColumns , tableRows	The table equivalent
5	string \ undefined	emptyMessage	Shown over an all-zero chart (workload: MSG-I007)
6	'card' \ 'maximized'	variant	card renders the Expand control (item 27); maximized renders larger and without it, inside ChartDialog

Dependencies: none beyond React.

Processing overview: 003_DD-SPD §6 (P-25). The workload chart plots order counts (quantities are in its table); the trend chart plots completed-order counts.

Return value: JSX (<section> containing <svg> , the disclosure button and the table).

6. dashboardFormat.ts

Field	Value
Description	Pure formatting functions for the dashboard's value mappings
Return type	strings
Created by / date	Claude / 2026-09-22
Last modified by / date	—

Preconditions: none.

Rule type: display mapping (003_BD §4).

Rule references

No	Rule / validator	Location	Notes
1	M-13, M-14, M-16, M-17, M-18, M-19	003_BD §4	One function per mapping

Arguments / return values

Function	Signature	Mapping
formatRate	(x: number, y: number) => string	M-13: <code>y = 0</code> → "—", else <code>Math.floor(100·x/y + 0.5)</code> + "%"
formatLeadTime	(averageDays: number \ null) => string	M-14: <code>null</code> → "—", else one decimal + 「日」, e.g. 「13.7日」
formatOrders , formatUnits	(n: number) => string	M-18: <code>formatNumber (Intl.NumberFormat('ja-JP') , src/lib/format.ts)</code> thousands separators + 「件」 / 「個」, e.g. 「1,234件」, 「17,922個」 (WI-005 DEC-006)
bucketLabel	(kind, weekStart, index) => string	M-16: 「納期遅れ」 (Overdue), 「今週」 (This week), the Monday as M/D (<code>monthDay</code> , e.g. 9/28), 「それ以降」 (Later)
weekRange	(start: string, end: string) => string	M-16: <code>YYYY/MM/DD~YYYY/MM/DD</code> for table equivalents
formatInZone	(utc: string, timeZone: string, part: 'dateTime' \ 'time') => string	M-17, M-20: <code>Intl.DateTimeFormat</code> with <code>timeZone</code> → <code>YYYY/MM/DD HH:mm</code> or <code>HH:mm:ss</code> ; the page wraps the first as 「[...] 時点 (Asia/Tokyo) 」
shownNote	(shown: number, total: number) => string \ null	M-19: 「{total}件中 {shown}件を表示」 (Showing {shown} of {total}) when <code>total > shown</code> , else <code>null</code>

Dependencies: none — no React, no DOM, so each is unit-tested directly.

Processing overview: the server sends counts and plant dates; this module turns them into text. Nothing here does calendar arithmetic. Every date it prints came from the server, so the browser's own time zone cannot shift a figure.

Return value: strings.

7. `dashboardApi.ts` — `getDashboard`

Field	Value
Description	Typed wrapper for <code>GET /api/dashboard</code>
Return type	<code>Promise<DashboardSnapshot></code>
Created by / date	Claude / 2026-09-22
Last modified by / date	—

Preconditions: none.

Not applicable — plain function.

Arguments

No	Type	Name	Description
1	AbortSignal	signal	Cancels the request on unmount

Dependencies: `ApiClient` (401 handling, JSON parsing, Problem Details mapping).

Processing overview: no transformation — the response type in `types.ts` mirrors 003_DD-API field for field.

Return value: the parsed snapshot, or an `ApiError` the page maps per 003_DD-SPD §2.

8. DashboardController.Get

Field	Value
Description	<code>GET /api/dashboard</code> — the request boundary
Return type	<code>ActionResult<DashboardResponse></code>
Created by / date	Claude / 2026-09-22
Last modified by / date	—

Preconditions: policy `ProductionOrderEditor` has authorized the request.

Not applicable — plain handler.

Arguments

No	Type	Name	Description
1	CancellationToken	cancellationToken	The only parameter; nothing is bound from the request

Dependencies

No	Module / component	Overview	Notes
1	<code>DashboardService.GetSnapshotAsync</code>	Use case	003_DD-FN §1

Processing overview: 003_DD-SPD §7. `[ResponseCache(NoStore = true, Location = ResponseCacheLocation.None)]` on the action.

Return value: 200 with the snapshot, or 401 / 403 / 500 as catalogued in 003_DD-API.

9. Server-side snapshot

Pointer only — `DashboardService`, `DashboardWindow`, `IDashboardReader` / `DashboardReader`, `DashboardMapper` and the `IPlantClock` additions are designed in **003_DD-FN**, which also specifies the span and the counter.

10. AppNavbar

Field	Value
Description	The navbar inside <code>AppHeader</code> on every authenticated screen (003_BD H-2, H-4, H-5; DEC-016)
Return type	JSX element
Created by / date	Claude / 2026-09-22
Last modified by / date	—

Preconditions: rendered by `AppHeader` only when a user is signed in.

Not applicable — plain component.

Arguments: none — the current route comes from `useLocation`.

Dependencies: `GuardedLink` (§11), `useLocation`.

Processing overview: 003_DD-SPD §8 (P-26). Current-entry rule: `/` → Dashboard; `/production-orders` and `/production-orders/{id}` → Production orders; `/production-orders/new` → New production order.

Return value: JSX (a `<nav>` named 「メインメニュー」 (Main menu), and on SP the 「メニュー」 (Menu) button and its panel).

11. NavigationGuard and GuardedLink

Field	Value
Description	A context through which an edited form can intercept in-app links, and the link component that consults it (DEC-022)
Return type	Context provider; hook; JSX link

Field	Value
Created by / date	Claude / 2026-09-22
Last modified by / date	—

Preconditions: `NavigationGuardProvider` wraps the authenticated routes in `App.tsx`.

Rule type: navigation interception.

Arguments / API

Member	Signature	Behavior
<code>useNavigationGuard().register</code>	<code>(guard: (to: string) => boolean) => () => void</code>	The form registers a function that returns <code>true</code> when it has taken over the navigation (it was dirty and opened its dialog). Returns the unregister function, called on unmount
<code>GuardedLink</code>	props of react-router <code>Link</code>	On click: if a registered guard returns <code>true</code> for <code>to</code> , <code>preventDefault()</code> ; otherwise a normal <code>Link</code> . Modified clicks (Ctrl/Cmd/Shift, middle button) are never intercepted: they open a new tab and lose nothing

Dependencies: react-router `Link` , `useNavigate` .

Processing overview: 003_DD-SPD §9 (P-27). Only one guard is registered at a time (one form per screen). The form's dialog remembers the destination and, on Discard, navigates there. This avoids migrating `BrowserRouter` to a data router for `useBlocker` . Browser Back, reload and closing the tab are intentionally not intercepted (DEC-022).

Return value: provider, hook, link.

12. `HealthIndicator` and `useSystemHealth`

Field	Value
Description	Item 26: shows HS-01–HS-05 with the last-check time; the hook owns the polling
Return type	JSX element; hook returning <code>{ server, database, checkedAt, state }</code>
Created by / date	Claude / 2026-09-22
Last modified by / date	—

Preconditions: rendered on SCR-003 outside the `forbidden` state.

Rule type: status derivation (003_BD HS-01–HS-05, DEC-020).

Arguments: none.

Dependencies: `systemApi.getHealth` (with a 5 s `AbortSignal.timeout`), `document.visibilityState`, `dashboardFormat.formatHealth` (M-20).

Processing overview: 003_DD-SPD §10 (P-28). The first check runs on mount; then every 30 s while the tab is visible; on becoming visible it checks at once. The polling uses `setTimeout` chained after each completed check, never `setInterval`, so a slow check cannot pile up concurrent requests.

Return value: JSX: a `role="status"` region with the two statuses, e.g. 「サーバー：正常」 「データベース：正常」, and 「HH:mm:ss に確認」 (Checked HH:mm:ss).

13. `ChartDialog`

Field	Value
Description	Items 28–29: a chart and its table in a full-window modal (DEC-018)
Return type	JSX element
Created by / date	Claude / 2026-09-22
Last modified by / date	—

Preconditions: opened by a chart's Expand control with that chart's props.

Not applicable — presentational.

Arguments: the same props as `BarChart` (§5), plus `open`, `onClose`, and `returnFocusTo` (the Expand button).

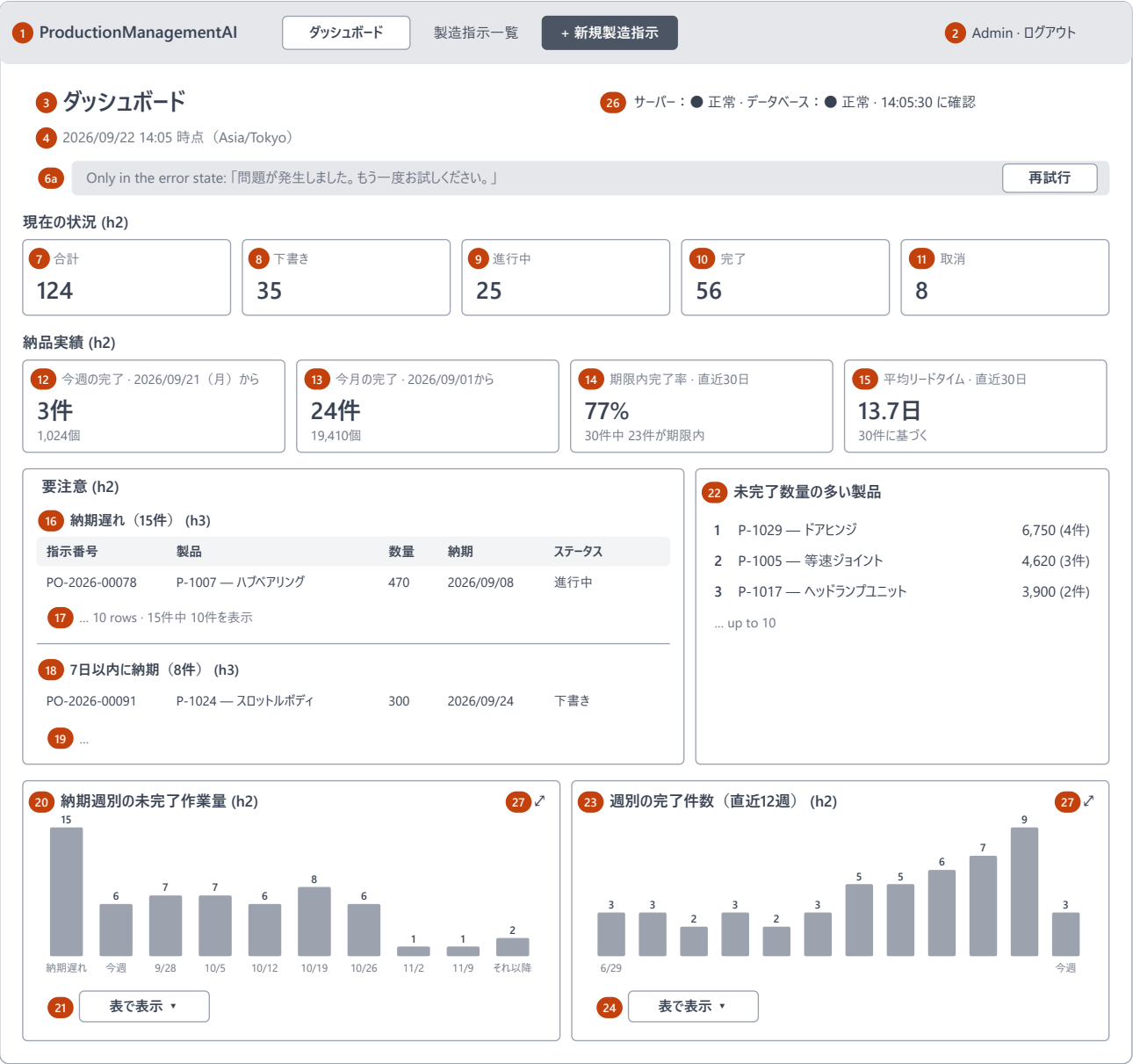
Dependencies: native `<dialog>` with `showModal()`; `BarChart` with `variant="maximized"`.

Processing overview: 003_DD-SPD §11 (P-29). The platform provides focus containment, Escape and the backdrop; the component sets `aria-labelledby` to the chart title and restores focus on close.

Return value: JSX (`<dialog>`).

Screen layout and mockup

Implementation-fidelity layout. It refines 003_BD §1 in three ways. The two tile rows get section labels (「現在の状況」 (Current state), 「納品実績」 (Delivery)) as `<h2>` headings, so the page has a heading outline. The delivery tiles state their windows in their captions. The two charts sit side by side at `1g` and stack below it.



Mockup artifact: <https://claude.ai/artifact/5f5hbKibAX3xURVAS5Aeot> — published privately on 2026-09-22 with the user's authorization ("public it privately"); version 2 republished to the same URL under plan revision 2 (step 7), version 3 with the Lucide icons (DEC-023). Source: docs/en/020_detailed-

design/003/mockups/003_DD_screen-c-mockup.html , which reuses 002_DD's token system and app-look CSS verbatim, so the three mockups read as one set. Its figures are computed from 003_DB's seed as it reads on a run day of 2026-09-22 (a Tuesday), not invented. Artboards (version 2 of the mockup, same URL): 1 ready (PC, with the navbar and a healthy indicator), 2 loading, 3 empty system, 4 no recent completions, 5 load error with the database unavailable, 6 forbidden, 7 SP layout with the menu closed, 8 SP menu open, 9 chart with its table, 10 maximized chart, 11 the navbar on SCR-002.

Japanese edition, with the page captions in Japanese too, linked from the Japanese PDF (WI-005): <https://claude.ai/artifact/BPRHgKEpaYBnZMCzCsUBJz> (private). Source: mockups/003_DD_screen-c-mockup.ja.html .

Region	Contains (field/control)	Notes
Header	AppHeader (items 1, 2)	Shared component, unchanged
Header	AppHeader with AppNavbar (003_BD H-1–H-5)	Shared by every screen; Menu on SP
Title row	Heading (3), snapshot time (4), health indicator (26)	Indicator right-aligned on PC, wraps below on SP
Banner	MessageBanner (6a)	<code>error</code> state only; <code>role="alert"</code> , carries 再試行 (Retry)
Current state	Items 7–11	<code><h2></code> + <code></code> of tiles
Delivery	Items 12–15	<code><h2></code> + <code></code> of tiles
Needs attention	Items 16–19	<code><h2></code> + two <code><h3></code> groups
Top products	Item 22	<code><h2></code> + <code></code>
Workload / Trend	Items 20–21 / 23–24, Expand (27)	<code><h2></code> + Expand + SVG + disclosure + table
Maximized chart	Items 28–29	<code><dialog></code> over the page
Loading	Item 25	Skeleton in each widget frame, <code>aria-busy</code> on the widget region

Screen item definition

SCR-003 has no input field (003_BD §5). Its only controls are the navbar (shared header), Retry, the two table disclosures, the two Expand controls and Restore; everything else is output rendered from `DashboardSnapshot` (003_DD-API).

Field	Type	Required	Validation rule	Source (BD ref)
Navbar links (×3), Menu (SP)	<code>GuardedLink</code> , <code>button</code>	—	none — navigation	003_BD H-2–H-5, E-29–E-31
Retry	<code>button</code>	—	none; present only in <code>error</code>	§3 item 6a, E-21

Field	Type	Required	Validation rule	Source (BD ref)
View as table (×2)	<code>button</code> with <code>aria-expanded</code>	—	none	§3 items 21, 24, E-23
Expand (×2)	<code>icon button</code> , name 「{chart title}を拡大」 (Expand {chart title})	—	none	§3 item 27, E-24
Restore	<code>button</code>	—	none; Escape equivalent	§3 item 29, E-25

Message catalog additions (`messages.ts` ; other UI strings are in its `labels` object, WI-005 DEC-007). 001_DD and 002_DD already define MSG-E001–MSG-E020 and MSG-I001–MSG-I004, so Screen C continues the same catalog:

ID	Text (Japanese, as implemented)	English gloss
MSG-E021	ダッシュボードを閲覧する権限がありません。	You don't have permission to view the dashboard.
MSG-I005	納期遅れの製造指示はありません。	No overdue orders.
MSG-I006	7日以内に納期の製造指示はありません。	No orders due in the next 7 days.
MSG-I007	未完了の製造指示はありません。	No open orders.
MSG-I008	直近30日に完了した製造指示はありません。	No orders completed in the last 30 days.

Reused unchanged: MSG-E013 (load failure).

Loading / empty / error / success states

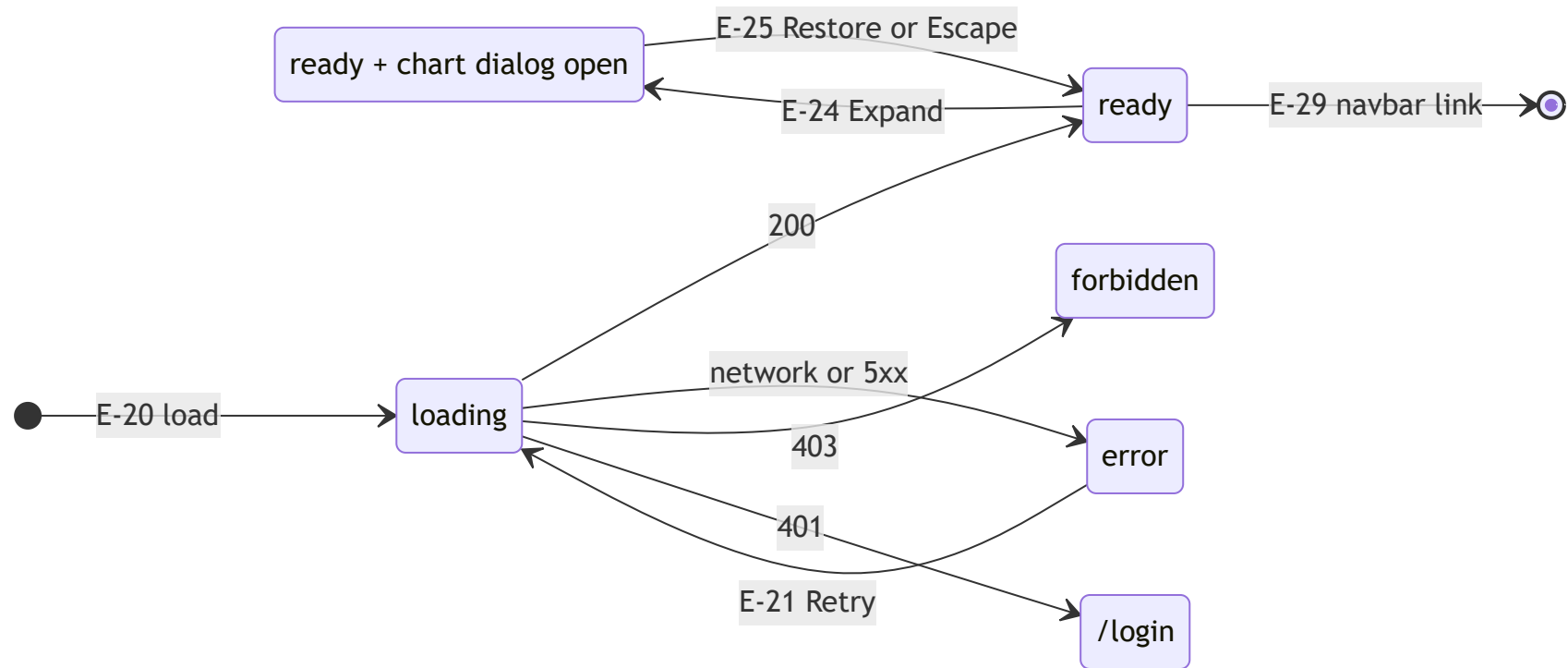
State	Trigger	UI behavior	Data shown
<code>loading</code>	Request in flight (mount or Retry)	Heading and page actions shown; each widget frame shows a skeleton; the widget region has <code>aria-busy="true"</code>	No figures
<code>ready</code>	200	Every widget from the one snapshot; snapshot time shown and announced once (「... 時点 (Asia/Tokyo) 」)	All figures
<code>ready</code> , empty system	200 with <code>statusCounts.total = 0</code>	Not a separate state: tiles show 0, groups show MSG-I005/MSG-I006, top products and workload show MSG-I007, on-time and lead time show "—" with MSG-I008, trend shows twelve zero bars	Zeros
<code>ready</code> , no recent completions	200 with <code>onTime.completedCount = 0</code>	Items 14 and 15 show "—" with MSG-I008; everything else as usual	—
<code>error</code>	Network failure or 5xx	MessageBanner MSG-E013 with 再試行 (Retry); no figures, no stale snapshot	—
<code>forbidden</code>	Client role gate or 403	Panel MSG-E021 in place of the widgets; navbar stays; no health polling	—
Health: checking	First check in flight	Indicator 「確認中...」 (Checking...)	—

State	Trigger	UI behavior	Data shown
Health: OK	Health 200 with <code>database: "ok"</code>	「サーバー：正常」・「データベース：正常」・「HH:mm:ss に確認」 (Server: OK · Database: OK · Checked HH:mm:ss)	—
Health: database unavailable	Health 200 with <code>database: "unavailable"</code>	"Server: OK · Database: Unavailable"; figures already shown stay	—
Health: server unreachable	No answer in 5 s, network failure or 5xx	「サーバー：接続不可」・「データベース：不明」 (Server: Unreachable · Database: Unknown)	—
Chart maximized	Expand	<code>ChartDialog</code> open over the page; the page behind is inert	Same snapshot
401	No valid session	<code>apiClient</code> redirects to <code>/login</code>	—

Processing and state transitions

This screen has no business state machine and changes no data (REQ-039). The one business transition this work item touches — `InProgress` → `Completed` recording its time — belongs to SCR-001 and is specified in 001_DD v4.

State transitions



Events that keep the state (E-23 View as table, E-26–E-28 health checks) and the return to / are in the table below, not the diagram.

From state	Event	To state	Side effect
(route entered)	E-20 load	loading	One GET /api/dashboard
loading	200	ready	Snapshot-time line announced
loading	401 / 403	/login / forbidden	—
loading	network or 5xx	error	—
error	E-21 Retry	loading	Same request re-sent
ready	E-23 View as table	ready	Table shown or hidden; no request

From state	Event	To state	Side effect
ready	E-24 Expand	ready + dialog open	No request; focus into the dialog
dialog open	E-25 Restore / Escape	ready	Focus back on Expand
any except forbidden	E-26–E-28 health check	same	Indicator updated; figures untouched
any	E-29 navbar link	(leaves screen)	Navigates to SCR-002 or SCR-001 create; health polling stops on unmount
(leaves)	returns to /	loading	A new snapshot; the earlier one is never reused (no-store)

Processing flows

Flow	Where (screen-processing-design / function-design section)	Events
P-20 Load	003_DD-SPD §1	E-20, E-21
P-21 Response handling	003_DD-SPD §2	E-20
P-22 Tiles	003_DD-SPD §3	—
P-23 Attention groups	003_DD-SPD §4	—
P-24 Top products	003_DD-SPD §5	—
P-25 Charts	003_DD-SPD §6	E-23
P-26 Navbar	003_DD-SPD §8	E-29–E-31
P-27 Navigation guard	003_DD-SPD §9	001_BD E-07a, E-09
P-28 Health polling	003_DD-SPD §10	E-26–E-28
P-29 Maximized chart	003_DD-SPD §11	E-24, E-25
Server snapshot	003_DD-FN §1–§5	—

APIs used

Endpoint	Method	Purpose	Design doc
/api/dashboard	GET	The snapshot	003_DD-API §1
/api/system/health	GET	Server and database status	003_DD-API §2

Database and transaction mapping

Operation	Table(s)	Transaction boundary	Concurrency handling
Dashboard snapshot	<code>production_orders</code> , joined to <code>products</code> for Q3 and Q4	One <code>REPEATABLE READ READ ONLY</code> transaction around seven statements (003_DB, DEC-015)	Not applicable — read-only; one snapshot for every figure
Record completion (SCR-001)	<code>production_orders.completed_at_utc</code>	The existing single <code>SaveChanges</code>	<code>xmin</code> token, unchanged (001_DD v4)
Health ping	none — <code>SELECT 1</code>	Single statement, 2 s timeout	Not applicable

Raw SQL is used for the snapshot's seven statements (003_DD-FN §3). That is a deliberate exception to the LINQ-first style of 001_DD and 002_DD, because their aggregate shapes have no clean LINQ translation. Each statement is a `FormattableString` , so every value is a bound parameter.

Exception handling

Failure	Retry policy	Timeout	User-facing error	Logging
Not signed in (401)	none	—	Redirect to <code>/login</code>	Existing <code>ApiClient</code> behavior
Missing role (403)	none	—	Panel MSG-E021	Information (framework)
Database or query failure (500)	none automatic; 再試行 (Retry) re-sends	Npgsql command timeout 30 s (default)	Banner MSG-E013	Error <code>DashboardSnapshotFailed</code> with the exception, server-side only; generic Problem Details to the client
Frontend network failure	none automatic; 再試行 (Retry)	browser default	Banner MSG-E013	<code>console.error</code> in development only
Unmount mid-request	not an error	—	none	none — aborted
Health: database ping fails or exceeds 2 s	none; next poll in 30 s	2 s (server)	Indicator "Database: Unavailable"	Warning <code>DatabasePingFailed</code> with the exception type only; counter <code>database=unavailable</code>
Health: server unreachable	none; next poll in 30 s	5 s (browser)	Indicator 「サーバー：接続不可」・「データベース：不明」 (Server: Unreachable・Database: Unknown)	none client-side
Health: 401	none	—	Redirect to <code>/login</code> (the poll did not renew the session, DEC-019)	Existing <code>ApiClient</code> behavior

Failure	Retry policy	Timeout	User-facing error	Logging
An old backend completes an order after 003_DB migration 1	not applicable at runtime	—	That save fails with 500	Prevented by deploy order (003_DB "Deploy order")

Accessibility (WCAG 2.2 AA)

Icons (DEC-023, 003_BD M-21) render through `components/icons.ts` with `aria-hidden="true"` and `focusable="false"`, 16 px, `currentColor`, always beside visible text or inside a control with its own accessible name.

Stated in 003_BD's non-functional requirements and made concrete in 003_DD-SPD §3–§6. In summary: one `<h1>`, then `<h2>` per widget and `<h3>` per attention group; tiles are list items with label before value; the attention tables have captions and `scope="col"` headers; each chart is `role="img"` with a summarizing name, prints every value as text, and offers a real table; emphasis never relies on color alone; the SP chart scroller is keyboard-focusable and labelled; the navbar is a labelled `<nav>` with `aria-current` on the current entry and an SP Menu button with `aria-expanded` that closes on Escape; the health indicator is a polite `role="status"` region that announces changes only, with statuses in text and shape-coded dots; the maximized chart is a native modal `<dialog>` named by its title, with focus returned to Expand on close; loading is `aria-busy`, errors `role="alert"`, the finished load a polite status message. Automated checks: `vitest-axe` on the page in each state, and `@axe-core/playwright` on the running screen (TC-217).

Test viewpoints and unresolved decisions

U = unit (xUnit Application/Domain, or Vitest + RTL), I = integration (`WebApplicationFactory` + Testcontainers Postgres), E = E2E (Playwright). These become TC-201 ... in TP-004 (plan revision 2).

Test data isolation. The integration database carries WI-003's and 003_DB's demo seed (003_DB "Environments"). Dashboard integration tests therefore run in their own test class with their own `IntegrationTestFixture` instance, and so their own container. Before each test they delete every production order through the fixture's owner connection (`ExecuteAsOwnerAsync`), insert exactly the rows the case needs, and pin the plant clock with `FakeTimeProvider`. TC-218, which checks the seed itself, runs in a separate class on a fresh container with the real clock, and deletes nothing.

Scenario	Precondition	Expected result	Test-plan ID
Landing page (REQ-028) — E	Seeded stack	Signing in lands on / showing the dashboard heading, every widget, the snapshot time and the navbar with Dashboard current; the placeholder home content is gone	TC-201

Scenario	Precondition	Expected result	Test-plan ID
Read-only (REQ-039, DEC-006) — U, E	Ready state	No tile, row, bar or list item is a link or has a click handler; the page issues no request other than <code>GET /api/dashboard</code> and the health checks; opening and closing a maximized chart issues none	TC-202
Status counts (REQ-029) — I, U(fe)	Rows in each status, and a status with none	Each count exact; the empty status is 0; <code>total</code> = sum; labels are Screen A's	TC-203
Overdue and due-soon boundaries (REQ-030, D-01, D-02) — I	T pinned; active orders due T - 1, T, T + 7, T + 8; completed and cancelled orders due T - 1	T - 1 overdue; T and T + 7 due soon; T + 8 in neither; terminal orders in neither; groups ordered by due date then order number; with 12 overdue, 10 returned and <code>total</code> = 12	TC-204
Workload buckets (REQ-031, D-03, DEC-009) — U, I	<code>DashboardWindow</code> for each weekday Mon-Sun; active orders at T - 1, T, W + 6, W + 7, W + 55, W + 56	Exactly 10 buckets; each order in exactly one; bucket boundaries per D-03 for every weekday; the current week's first date is T; counts sum to draft + inProgress; empty buckets 0	TC-205
Top products (REQ-032, D-04) — I	12 products with active quantity, two tied, plus a product with only completed orders	10 returned, quantity descending, tie broken by SKU; the completed-only product absent	TC-206
Completion recorded on save (REQ-033) — U, I	InProgress order; clock pinned	Save to <code>Completed</code> sets <code>completed_at_utc</code> = save time = <code>updated_at_utc</code> ; a later notes edit leaves it unchanged; other transitions never set it; a 409 or 422 save sets nothing; a body with <code>completedAtUtc</code> is ignored	TC-207
Completion constraints and backfill (REQ-033, 003_DB) — I	Migrated database	Owner <code>UPDATE</code> making a <code>Completed</code> row's <code>completed_at_utc</code> NULL, or a <code>Draft</code> row's non-NULL, or <code>completed_at_utc</code> < <code>created_at_utc</code> → check violation; after migration every <code>Completed</code> row has a value; <code>pmai_app</code> can still update orders and still cannot <code>DELETE</code>	TC-208
Completed this week / month (REQ-034, D-05, D-06) — U, I	T = a Wednesday; completions at Monday 00:00 JST (= Sunday 15:00 UTC), Sunday 23:59 JST the week before, the 1st 00:00 JST, the last day of the previous month	Week and month counts include exactly the in-window ones; plant-time boundaries, not UTC; quantities summed; no completions → 0 / 0	TC-209
On-time rate (REQ-035, D-07) — U, I	Completions on the due date, one day late, early; one at T - 29 and one at T - 30; a cancelled order	On the due date counts as on time; late does not; T - 29 in the window, T - 30 out; cancelled in neither count; client: <code>23/30</code> → "77%", <code>1/3</code> → "33%", <code>2/3</code> → "67%", <code>0/0</code> → "—" + MSG-I008	TC-210
Completion trend (REQ-036, D-08) — U, I	Completions in 3 of the 12 weeks, one at a week boundary in plant time	Exactly 12 entries oldest first; the 9 empty weeks are 0; the boundary completion lands in the plant-local week; a completion 13 weeks ago is excluded	TC-211

Scenario	Precondition	Expected result	Test-plan ID
Average lead time (REQ-037, D-09) — U, I	Three orders with lead times 1.25, 2.0 and 2.1 days	<code>averageDays</code> = 1.8 (mean 1.783... rounded); none in the window → <code>null</code> → "—"; client prints "1.8 days"	TC-212
Authorization (REQ-038) — I, E	—	Unauthenticated → 401 and the UI redirects to <code>/login</code> ; signed in without either role → 403, no figures in the body, panel MSG-E021	TC-213
One snapshot (REQ-028, DEC-011, DEC-015) — I	Mixed data	Workload counts sum to draft + inProgress; overdue bucket = overdue <code>total</code> ; the reader marks its transaction read-only before its first query — observed through Npgsql's own statement tracing (<code>SET TRANSACTION READ ONLY</code> precedes all seven <code>production_orders</code> statements), since the reader's ADO.NET commands are invisible to EF interceptors (DEC-024); the <code>REPEATABLE READ</code> isolation is set by the reader's <code>BeginTransactionAsync</code> call and reviewed in code	TC-214
Empty system (REQ-028) — I, U(fe)	No orders	200; all zeros; <code>averageDays</code> null; client shows MSG-I005–MSG-I008 and "—" per 003_DD states; no error	TC-215
Load failure and Retry (REQ-028) — U(fe), E	API returns 500	Banner MSG-E013 with Retry , no figures; page actions usable; Retry re-requests and renders on success	TC-216
Icons decorative (DEC-023) — U	Each component with icons	Every Lucide <code><svg></code> has <code>aria-hidden="true"</code> ; icon-only controls have an accessible name; no text is replaced by an icon	TC-228
Accessibility — U (axe), E (axe)	Each state	No axe violations; heading outline h1 → h2 → h3; each chart <code>role="img"</code> with a name listing every bar's value; View as table toggles <code>aria-expanded</code> and reveals a table with the same numbers; SP chart scroller focusable	TC-217
Demo seed figures (003_DB) — I	Fresh container, real clock, migrations only	124 orders (35/25/56/8); every one of the 12 trend weeks ≥ 2; on time 23 of 30; lead time 13.7 over 30; every workload bucket non-zero; 10 top products; no negative lead time; next order number continues after the new seed	TC-218
Index usage (003_DB) — I	Seeded data, <code>ANALYZE</code>	<code>EXPLAIN</code> of Q3a uses <code>ix_production_orders_active_due_date</code> ; Q5 uses <code>ix_production_orders_completed_at_utc</code>	TC-219
SP layout (003_BD SP) — E	Pixel 7 profile	Tiles two per row; attention rows as cards; chart cards scroll horizontally, the page does not	TC-220
Query string ignored (003_DD-API) — I	—	<code>GET /api/dashboard?today=2020-01-01&top=1000</code> returns the same body as without the query string	TC-221

| Navbar (REQ-040, DEC-016) — U, E | Signed in on each of `/`, `/production-orders`, `/production-orders/{id}`, `/production-orders/new`; Pixel 7 profile | Three links on every screen; `aria-current="page"` on the right one per 003_BD H-2; not rendered on `/login`; SP: Menu toggles `aria-expanded`, Escape closes and

returns focus; each link navigates | TC-222 | | Navigation guard (REQ-019, DEC-022) — U, E | SCR-001 with an edited field | Navbar, breadcrumb and app-name links open the discard dialog; Keep editing stays with values kept; Discard goes to the clicked link's destination; with no edits each link navigates at once; a Ctrl-click is not intercepted | TC-223 | | Health endpoint (REQ-041, DEC-020) — I | App as `pmai_app` | Unauthenticated 401; no role 403; healthy → 200 { `database: "ok", checkedAt` } and nothing else; with `IDatabasePing` replaced to throw or to exceed 2 s → 200 { `database: "unavailable"` }; `Cache-Control: no-store`; no exception text in the body | TC-224 | | Health poll never renews the session (DEC-019) — I | Signed in; cookie clock advanced past half its lifetime | `GET /api/system/health` returns no `Set-Cookie`; `GET /api/dashboard` at the same moment does (renewal still works for real use) | TC-225 | | Health indicator (REQ-041) — U(fe) | Fake timers; mocked `getHealth` | 「確認中…」 (Checking...) then OK; `unavailable` → "Database: Unavailable"; rejection or 5xx → "Server: Unreachable · Database: Unknown"; next check after 30 s, chained (no overlap); hidden tab pauses, visible tab checks at once; the live region announces a change but not a repeat; unmount stops polling | TC-226 | | Maximize a chart (REQ-042, DEC-018) — U(fe), E | Ready state | Expand opens a modal dialog named by the chart title, showing the chart and its table with the same numbers; no network request; Tab stays inside; Escape and Restore close it and focus returns to Expand; axe clean with the dialog open | TC-227 |

Regression in WI-003's own suites is a plan revision 3 task, not a new viewpoint. The tests that assert the 80-row total, the 32/24/16/8 status spread or the `00081` next number move to 003_DB's figures. The tests that assert relative dates do not change.

Unresolved decisions: none for the design. The same two limitations as 002_DD carry over:

- The dashboard reuses the `ProductionOrderEditor` policy, so a read-only role cannot be expressed; this waits on WI-001 DEC-015.
- The mockup is a static rendering: it fixes layout, state coverage, and the treatment of tiles, groups and charts. Behavior is specified in 003_DD-SPD.